



WEBB

EXE

ROBOTICS

CHASING INNOVATION

We believe in developing innovative, high-performing robots through effective collaboration while fostering collaborative engineering environments



JERRY

Co-Captain, Lead Mechanic



LINDA

Co-Captain, Lead Programmer



JONATHAN

Mechanic, Builder



JAKE

Mechanic, Builder



TINA

Mechanic, Outreach



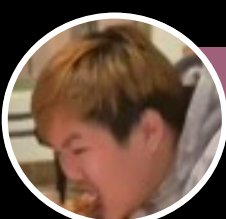
DAVID

Mechanic, Human Player



MAX

Programmer



TYLER

Driving Coach, Game Strategy



BOYANG

Lead Outreach



ABOUT US

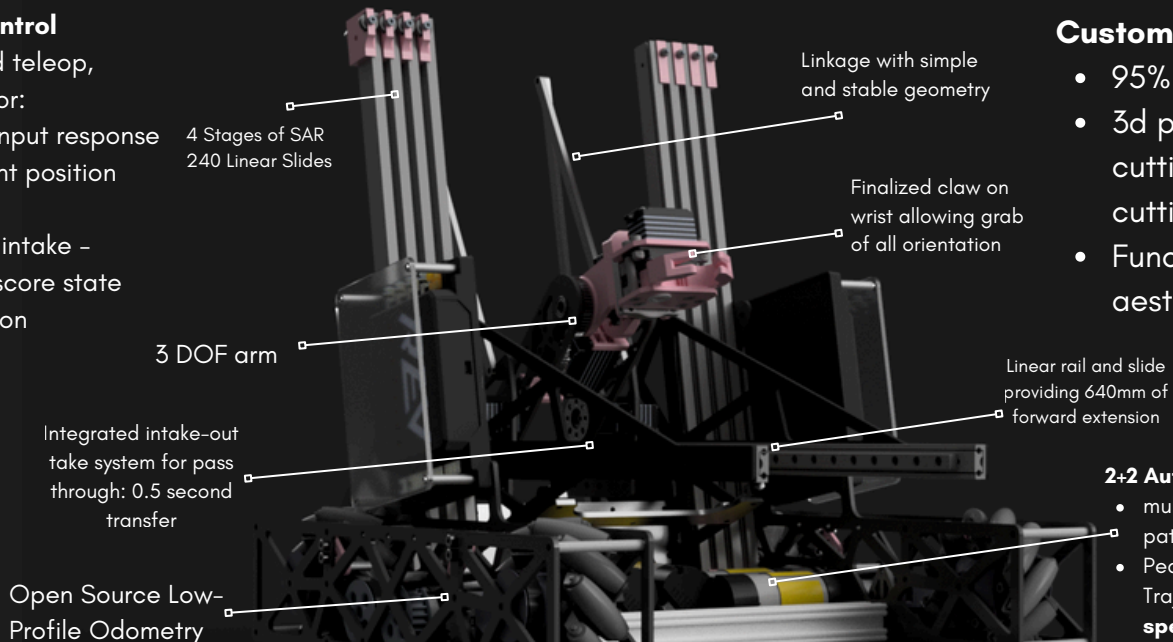
We are FTC Team #359 "WEBB.EXE" from The Webb Schools in Claremont, California. Founded in 2007, our mission is to develop, inspire, and educate, promoting a culture of innovation and a passion for learning in our community and beyond throughout our participation in FTC. We strive to empower students from diverse backgrounds, nurturing a prospering environment where every member can thrive and contribute.

Participating in FTC challenges us to think critically and work collaboratively, pushing the boundaries of what we can achieve together. Our journey is not just about building robots; it's about fostering the leaders, thinkers, and innovators of tomorrow. We are committed to upholding the principles of FIRST, adhering to and promoting gracious professionalism, teamwork, and the pursuit of excellence in all our endeavors. For us, FIRST is not just a learning process; it's about building a community united by the shared goal of making a positive impact on the world through science and technology.

Intelligent Control

Multi-threaded teleop, which allows for:

- Accurate input response
- Precise joint position timings
- Advanced intake - transfer - score state manipulation



Custom Parts

- 95% custom parts
- 3d printing, laser cutting, plasma cutting, and CNC
- Functional and aesthetic design

2+2 Autonomous

- multithreaded autonomous pathing library
- Pedro Pathing Based Trajectory for **3+0 specimen auto** and **4+0 Sample auto** adapting to alliance strategies.

OUTREACH HIGHLIGHTS

- Hosted all SoCal League E meets at our school
- Connected with 7 FTC teams, 2 VRC teams, and 1 FLL team
- Met and collaborated with a Marine Biology Researcher, Mercedes Systems Engineer, and Harvey Mudd College Engineering and CS Professors
- Invited to attend the Rockwell automation fair as representative for the FTC

SEASON HIGHLIGHTS

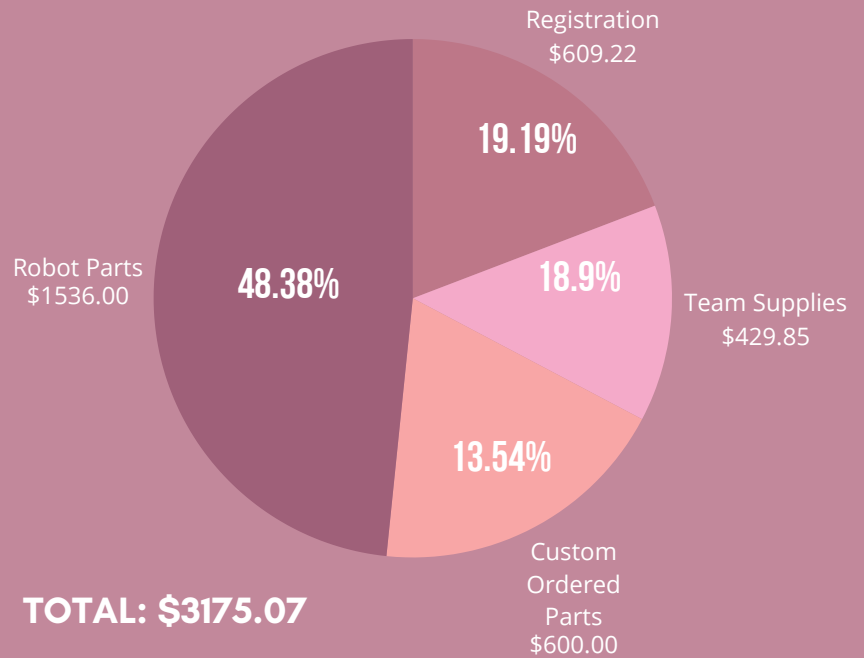
- League rank 1 in SoCal League E
- Three specimens 63pt autonomous
- Four sample 67 pt autonomous
- Average 10 sample teleop
- 150pt solo score

Season Plan & Budget

FUNDING & BUDGET

We are immensely grateful to The Webb Schools, our school, which fostered our robotics journey by generously furnishing us with practical resources and welcoming learning spaces. Additionally, our team is privileged to be financially sponsored by Webb as well, whose benevolent support made BOT3 possible in regards to securing essential robot components and custom ordered CNC parts. The right image projects a breakdown of our season's expenses, categorized to reflect our financial allocations.

BUDGET ALLOCATION



SEASON SCHEDULE

PLANNING, EXECUTION, IMPORTANT DATES



OUR SPONSORS



ROCKWELL AUTOMATION

Our team had the incredible opportunity to present our robot at the 2024 Rockwell Automation Fair. This event allowed us to introduce FIRST Tech Challenge (FTC) and this year's game to a diverse audience of engineers and industry professionals. Beyond showcasing our own work, we were inspired by cutting-edge automation technologies and engaged in valuable discussions with professionals at the forefront of industrial innovation. This experience reinforced the real-world applications of robotics and engineering.



LEAGUE CONNECTIONS

We are a longstanding host of the Southern California League E. This year, we hosted Meets 0, 1, 2, and 3 at the Webb Schools. Apart from volunteering for setting and cleaning up, parking management, referee, MC, and more, we also actively engaged with our local community, joining scrimmages and inviting local teams to visit our lab.



SEASON GOALS

- Expanding outreach efforts to alumni
- Giving back to the FTC community
- Connecting with our league
- Develop outstanding robot-building skills
- Assist other teams in improving their engineering and design skills

COMMUNITY CONNECTIONS

Our team is dedicated to expanding the robotics program to our community. We reached out to the Webb admissions department to allow us to connect with prospective students and visited nearby colleges to advance our engineering and programming skills.

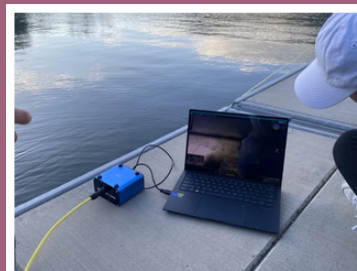
SUBMARINE PROJECT

In the fall of 2024, we reached out to the Raymond M. Alf Museum of Paleontology and started a collaboration project related to under water fossil excavations in Turlock Lake. Through extensive process of R&D we put together a manually controlled submarine that can perform under water mapping and fossil locating.



The ROV is equipped with 8 thrusters, allowing 6-degree-of-freedom movement. It features a depth rating of up to 100 meters, supported by a 300-meter tether cable. The onboard HD camera provides real-time imagery with bright 1500 lumen LEDs for improved visibility. Navigation and control are managed through ArduSub an open source software system, enabling autonomous pathing and real time control.

Through this experience combining expertise in mechanical engineering, robotics, and paleontology, we not only honed our technical skills but also developed a deeper appreciation for the ecological and historical significance of underwater environments. This collaboration sets the stage for future innovations in both fossil excavation and ocean exploration.



SUPPORTED TEAMS

FTC Team 358 & FTC Team 19888 (League rank 6)

We mentored our sibling teams, FTC 358 and FTC 19888, and also an up-and-rising rookie team, guiding them in design, CAD, and strategy throughout the season. Through collaboration and regular scrimmages, we helped refine their engineering process and foster innovation.

FTC Team 11347 CircuitRunners Robotics - Blackout

We mentored team 11347, who asked for our advice on their intake mechanism after seeing one of our earlier designs that we posted on the FTC Discord server. We gave them feedback on their current designs and inspired them to develop an intake mechanism with increased efficiency in scoring specimens.

FTC 24748 Team Atom

Throughout the season we guided FTC Team 24748, a rookie team from Kazakhstan. Our support focused on CAD and the development of their intake mechanism, helping them refine their design process. Additionally, we introduced them to Fusion 360, equipping them with the skills to create and iterate on their robot designs.

FTC Team 19013 G-Force

An FTC Team based in India that we helped in their designing process. We understand the challenge of starting on advanced CAD and assembly design, thus helping them by exchanging ideas and sharing design principles through online seminars in refining their engineering approaches.

FTC Team 23929 GrizzlyTech:

We donated out-of-stock parts to GrizzlyTech and also reached out to Pratush, their lead programmer, to assist in their autonomous pathing and tuning. GrizzlyTech also visited our lab, where we demonstrated how we used a laser cutter to quickly manufacture custom parts.

FLL Team 60444 Slipper Guys

We mentored Slipper Guys while on break, teaching more advanced code structures and helping them set design goals for their individual subsystems. We also gave them feedback on their current module power delivery interface and how to improve it.

VRC Team 7271IP Σ^3

A private student-run Vex Team from The Webb Schools. We collaborated with them to expand our programming horizons, developing an autonomous PID Library in both Java and C++. Members of our team have also tagged along their Signature Events: One World Showcase Event @ UC Berkeley and NorCal Silicon Valley Signature Event and the Rumble In the Jungle Tournament, which they went undefeated in using the PID library we developed in together.

VRC Team 7700V Rolling Robots

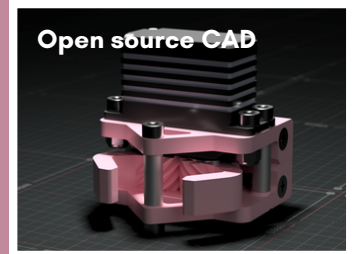
A rookie middle school VEX Team based in Rolling Hills, CA. Our team mentored them by improving their interview, programming, and design. Additionally, we also aided them in tuning their autonomous programs.

OUTREACH SNAPSHOTS

Aided G-Force with design process



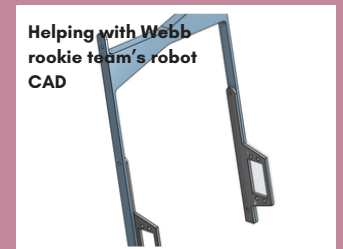
Slipper Guys



Open source CAD



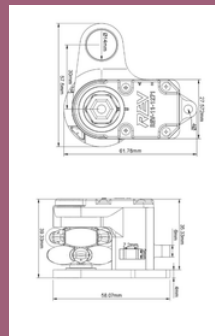
Mentoring
Webb rookie
team



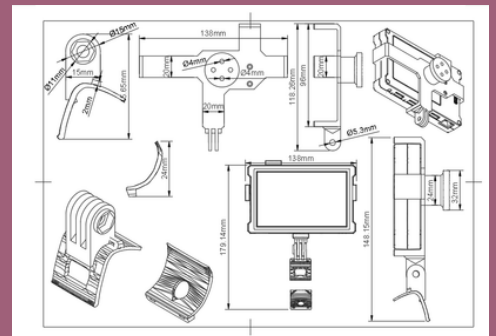
Helping with Webb
rookie team's robot
CAD

GIVING BACK TO THE COMMUNITY

We published online open-source odometry pods, TinyOdo, and our controller hub mount. TinyOdo combines high functionality with a compact design, optimized for a minimal turning radius on any chassis. The controller hub mount offers a secure, easy-to-install solution with zip-tie compatibility. Both highlight our commitment to open-source CAD, emphasizing accessibility, innovation, and quality in advancing technology.



TinyOdo



Driver Hub mount

OUR DESIGN PROCESS

Having a planned design process is crucial for efficiently designing a high-performance robot. It allows us to systematically approach our design while considering all necessary factors and constraints.

1. Define the Problem

After developing a thorough understanding of the game, we define the task that the part or subsystem must accomplish.

2. Specify Requirements.

The objective of this step is to identify and define all constraints comprehensively. This ensures that potential issues during the implementation phase are minimized or avoided altogether.

- Are there any size or weight limitations for this part/subsystem?
- If applicable, how many motors or servos can it use?

3. Brainstorm ideas

Look back at previous seasons for potential inspirations on similar tasks such as intakes and lifts. Research real-world problems that are similar and their solutions. Conducting research on real-world problems and their solutions, such as car suspension systems or crane arms, can provide valuable insights for optimizing the robot's design and functionality in the current robotics competition.

4. Develop a preliminary design and quickly fabricate it

Nothing fancy is needed in CAD besides a part that can be quickly manufactured using processes like laser cutting and 3d printing or built from standard parts. The prototype's purpose is not to be final but to enable us to test and determine its viability before committing further.

5. Test prototype

We test for:

- Effectiveness: does it solve the problem it was designed to?
- Weaknesses: what are the problems with the design?
- Practicality & Feasibility: similar to effectiveness. Is the design overcomplicated? Hard to manufacture?

6. Analyze Results

After testing, we reevaluate the design (how well can we optimize it?). Is it worth pursuing, or should we try a different approach?

7. Go back to step #4 until we are ready to move to step #8

We continue the prototyping and testing cycle until we feel that we have a design that we are confident in. We should have already improved it significantly from the initial prototype and should have learned enough from testing and analysis to make a final implementation.

8. Final Implementation in CAD

After improving the initial prototype and learning from testing and analysis, we can proceed with the final implementation. See the CAD organization page for how we keep track of our whole assembly.

BREAKING DOWN THE GAME (9.20.24)

We watched the kickoff video and read Game Manual Part 2 to understand the game and scoring tasks. Using cost-benefit analysis, we prioritized tasks and planned subsystem development.

Key Functional Requirements (in order of R&D importance):

Robot is able to score both specimens and samples, extend into the submersible and be able to pick out the right color samples. Additionally is able to easily intake game elements from the human player station, both on the floor and grabbing from the wall.

Subsystems required to meet Key Functional Requirements:

Drivetrain, lift system, intake, extension mechanism

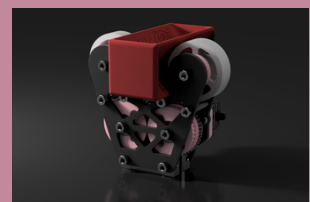
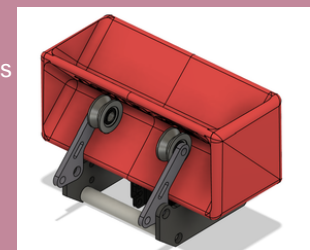
BRAINSTORMING (9.20.24)

After understanding the game and design needs, we brainstormed, drawing from past robots, choosing a horizontal extension crane design with a passthrough mechanism. We also explored intake options like a compliant wheel and traditional claws.



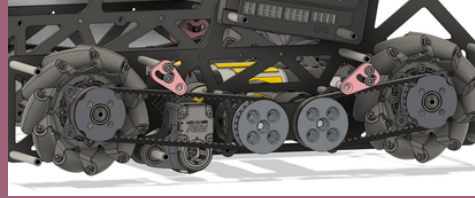
PROTOTYPING (9.22.24)

After brainstorming, we created CAD models of potential designs for testing in our next work session. Our focus was prototyping intake mechanisms, as their performance would heavily impact other systems. Through testing, we selected a traditional claw design. While slower than an active intake, it proved more reliable, robust, and easier to integrate into our autonomous path, meeting our functional and design goals.

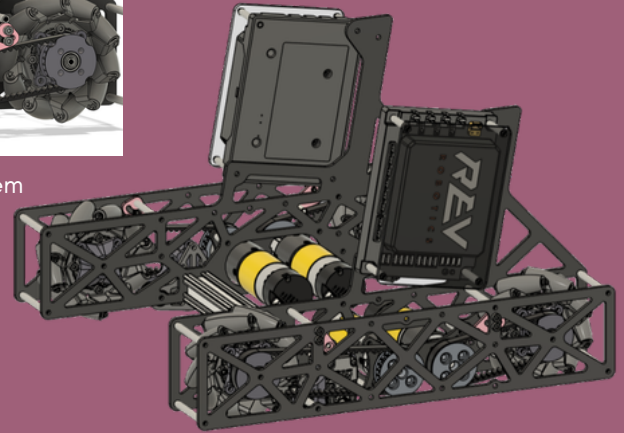


SUBSYSTEM OVERVIEW

- The chassis uses parallel plate and belt-driven design
- Using 435rpm motors and a 1:1 pulley ratio, the max speed of the drivetrain is 0.696 m/s
- Belt-drive allows for smooth power transmission and minimal maintenance
- This chassis serves as a base for the other modules of the robot: the intake, arm, horizontal extension, and lift
- To reduce backlash, we designed adjustable belt tensioners



The belt driven system

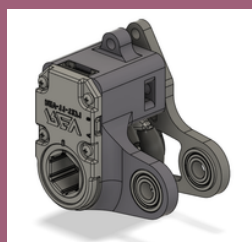
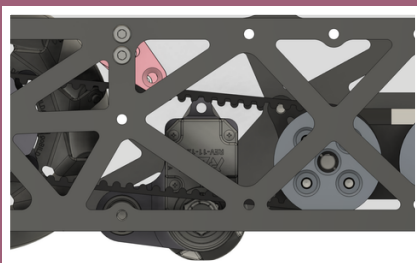
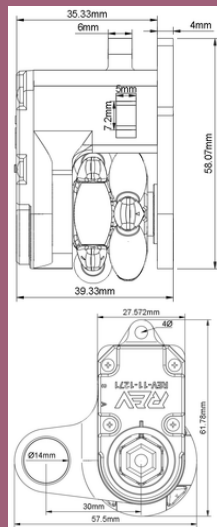


FUNCTIONAL REQUIREMENTS

- Integration of all the subsystem mounting into 4 parallel plates, reducing excess weight
- Consistent and adjustable belt tensioning to prevent backlash
- Lightweight and aesthetic design

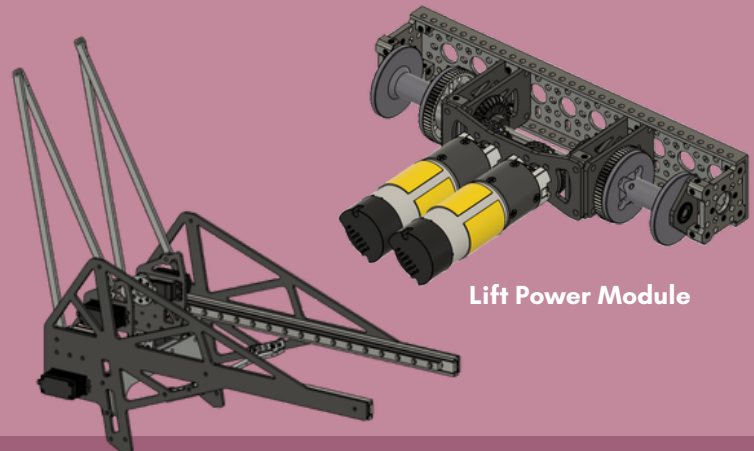
ODOMETRY

Leveraging open-source innovation, our team has engineered cost-effective odometry pods "TinyOdo" (Figure Right), priced competitively at \$65, which provide double the Clicks Per Rotation (CPR) compared to GoBilda's \$100 alternatives. Adhering to design requirements, our pods boast a smaller turning radius, surpassing other models in agility and performance. Central to our navigation system is the implementation of 3-wheel odometry, offering enhanced accuracy and control (Figure Bottom). This aligns seamlessly with our robot's overarching architecture.



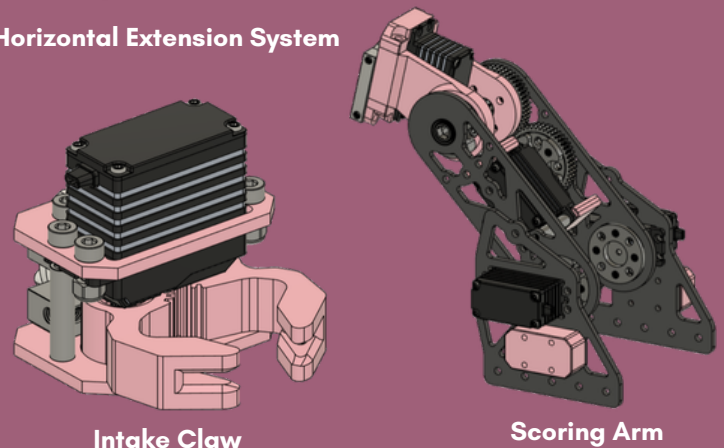
Subsystem Integration

Each subsystem is designed for quick removal and reattachment to the drivetrain, simplifying modifications. The chassis includes mounting holes for every subsystem and accommodates additional modules.



Lift Power Module

Horizontal Extension System



Intake Claw

Scoring Arm

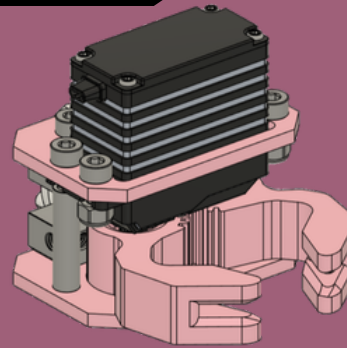
FUNCTIONAL REQUIREMENTS

- Quick and efficient intake off the ground
- The feature of intaking specimen from wall
- The ability to intake from any orientation
- Lightweight and easily maintainable design
- Durability to withstand heavy use and potential robot collisions

LESSONS LEARNED

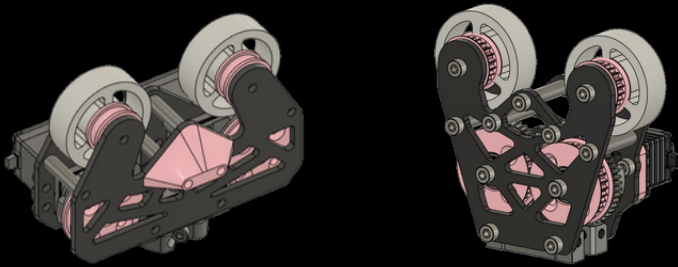
Our intake process is highly efficient at picking up the sample from the play field. However, with inconsistency issues that we encountered last year when transferring the pixels from the intake into the scoring arm, we chose to use a passthrough mechanism this year, removing the need for any transfer.

SUBSYSTEM OVERVIEW



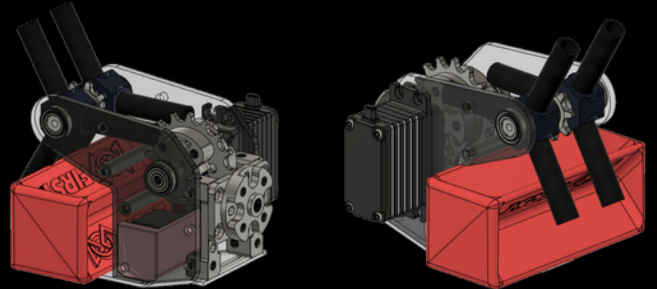
- Compact and lightweight geared arm design
- Up to 30° of error in rotation of the sample
- Ability to grab the sample from the edge
- Two Plate modular design for easy maintenance
- Utilizes one 25kg GoBilda Torque servo

V1 Design



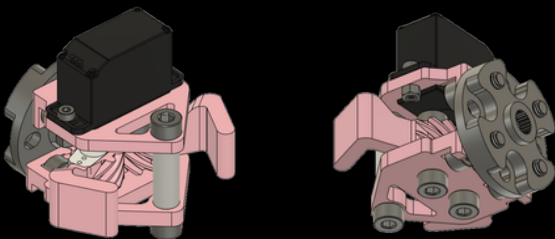
Our first prototype of an intake design involved using 2 compliant wheels that compressed the sample into a custom alignment geometry. Our first iteration featured two servos, powering two 2.5-inch compliant wheels. Our second iteration improved on this design reducing to single servo, reducing weight. However, through testing, the system is unreliable due to inconsistencies in servo power, resulting in a lack of grip in many intake situations.

V2 Design



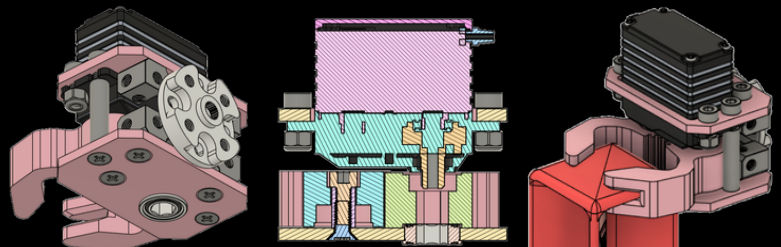
Our second approach to intake design was an active intake design involving surgical tubing. From our previous year in centerstage, we knew that surgical tubing would give us the fastest intake speed. We developed designs using geared-up servos and a bare 6000-rpm motor. However with both designs, although having the ability to intake quickly, we had issues with integrating it into our passthrough system that have limited space.

V3 Design



Through our testing and design of multiple active intake systems we realized that switching to a simpler, and more lightweight traditional claw would work most reliably. With our size constraints and the benefits of a light design we developed multiple iterations of a single servo geared arm claw. We made iterations for micro 9kg servos and standard 25kg servos. The design also features the alignment geometry used in our initial prototyping.

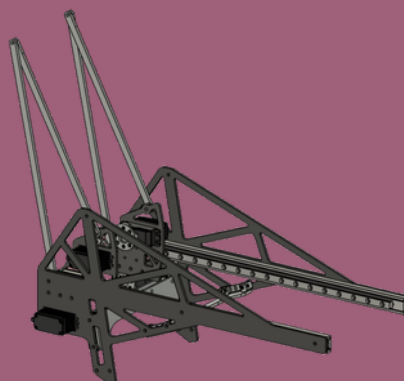
Current Design



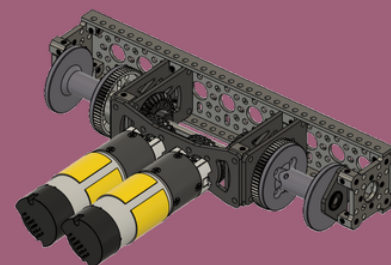
Our current intake system is finalized on a single 25kg servo powering 2 geared arms in a simple, easily maintainable two-plate design. The arm's geometry is specially designed to intake with 30° margin of error, and the ability to intake samples from a cluster. The geared arms have a 2 prong design, which allows for samples to be manipulated even when only the edge is contacted.

SUBSYSTEM OVERVIEW

- Two Extension Systems: Our robot uses a vertical lift system and a horizontal extension mechanism.
- Vertical Lift: Raises and lowers the robot to allow the lift assembly to score high bar specimens and high bucket samples.
 - powered by two 1150rpm GoBuilda motors on a 16mm spool, resulting in a 0.306m/s maximum lift speed
- Horizontal Extension: reduces drive distance during cycles and enables intake to reach in the submersible zone
 - powered by 2 Axon MAX Servos connected to linkages
 - allows for precise control of the extension position



Horizontal Extension System



Vertical Lift Power Module

FUNCTIONAL REQUIREMENTS

Horizontal Extension

- precise control of extension position
- reach deep into submersible
- pass through robot to deposit specimens and samples

Vertical Extension

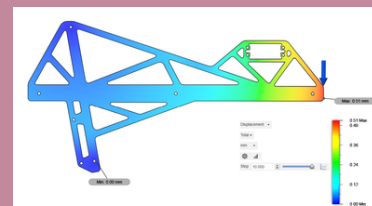
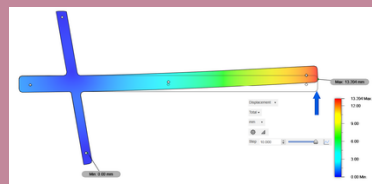
- strong and stable to support entire horizontal extension subsystem
- able to reach top bucket

LESSONS LEARNED

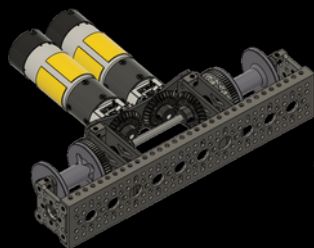
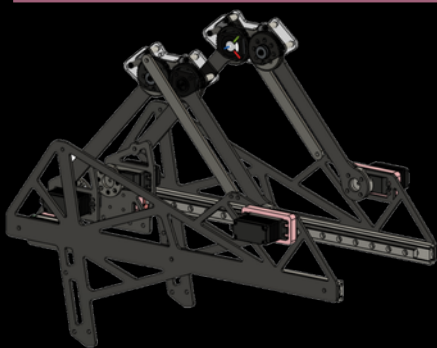
When stringing the lift slides, we learned that a counter string, which pulls the lift in is necessary. The lift power module has 2 spools for this reason. One spool is wrapped clockwise pulls the lift up. The other is wrapped counterclockwise and pulls the lift down.

System Optimization

Instead of the traditional method of separating horizontal and vertical extension, our two extension systems are integrated, with the horizontal extension mounted onto the vertical slides with a side plate. Tested through finite element analysis simulations, we settled on a geometry that can withstand the motions of the linear rail. As shown in strain analysis, with extra bracing and support the side plate withstands force much better at the tip. All servos are concentrately placed for the best center of mass and sub-system rigidity.

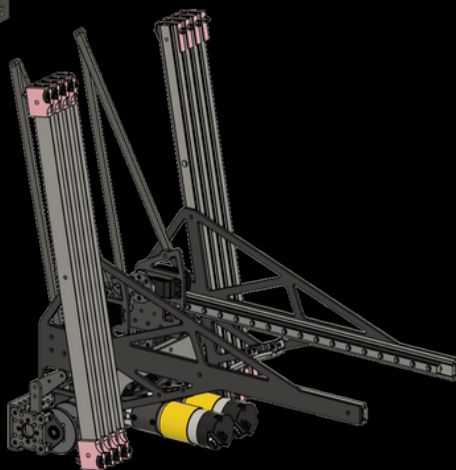


V.1 DESIGN



- The initial design was a geared double reverse 4-bar linkage powered by 2 servos mounted near the front of the robot
- It allowed for maximum possible extension range from a servo powered linkage
- The side plates serve as the attachment point for both the vertical slides and horizontal extension components
- The limitations of this design were:
 - 1. the stability and reliability of the complex linkage system
 - 2. the excess strain put on the servos from the large mechanical disadvantage

CURRENT DESIGN



- The extension servo mounts were moved to the lower back of the lift side plates, concentrating the center of gravity of the robot and allowing for a simpler linkage design
- The horizontal extension was changed to a simple two-stage linkage
- This new approach greatly reduces servo strain and structural complexity, with only a slight reduction in maximum reach
- The vertical extensions are still counterstrung, eliminating the risk of lift jams
- The lift now uses thick 2mm diameter kevlar string rather than thin fishing line for extra reliability due to lower risk of jams

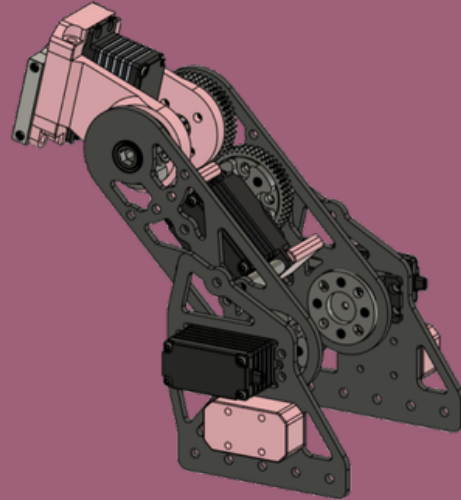
FUNCTIONAL REQUIREMENTS

- Many degrees of freedom necessary:
 - Main joint to reach into submersible
 - Wrist joint to pick up sample from ground and specimen from wall
 - Pivot joint to rotate specimens after passthrough to score

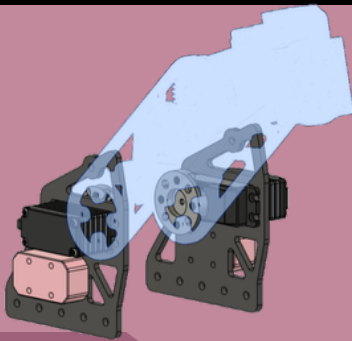
LESSONS LEARNED

With so many servos and moving parts in this subsystem, wiring was a major challenge. Though the design of the arm itself only went through 1 iteration, our cable management has drastically improved. Now every cable is shielded, and we use hard plastic to support the cable where it connects to the servo.

SUBSYSTEM OVERVIEW

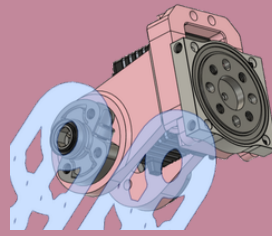


- 3 Degrees of freedom
- Powered by 4 Axon Max Servos
- Responsible for moving the intake claw to reach into the submersible, score specimens, and score samples



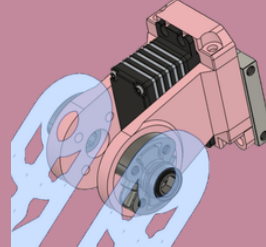
Base Joint

The base joint is powered by 2 Axon MAX servos that are mounted to the linear using 2 aluminum plates. These plates also attach to the extension linkage



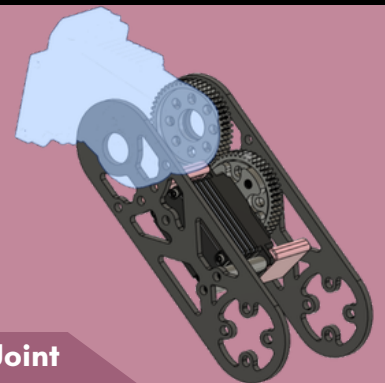
Pivot Joint

The pivot joint is powered by 1 Axon MAX servo, and is supported with a GoBuilda 32mm ID Bearing. The extra support on the servo horn is necessary due to the force experienced by the intake claw attached to this when scoring specimens.



Wrist Joint

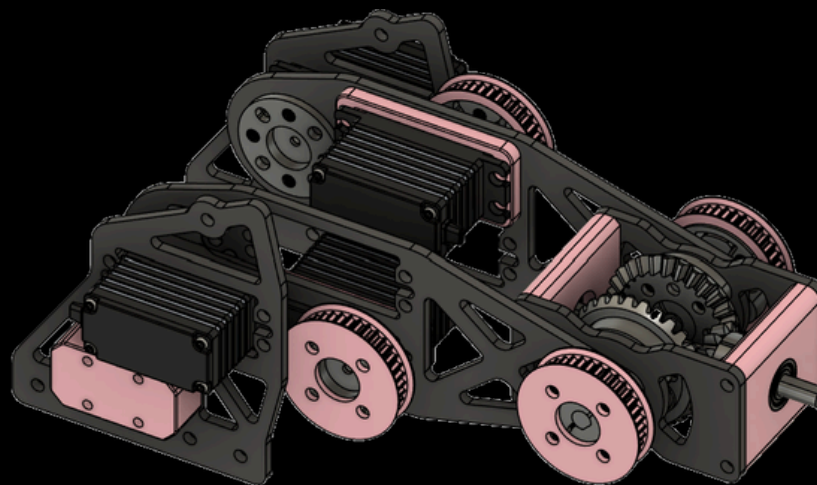
The wrist joint is also powered by 1 Axon MAX servo. The power is transferred by a 1:1 gear ratio, which is used to position the servo inside the arm for a more compact arm. The overall arm uses parallel plate construction much like the rest of the robot.



V.1 DESIGN

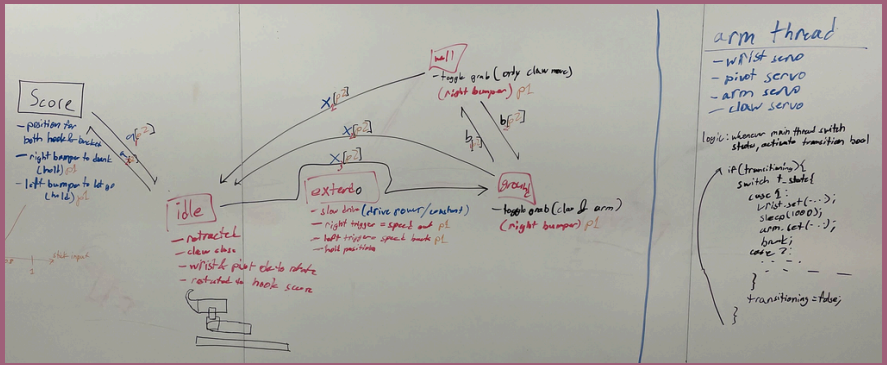
The original design of this arm also had 3 degrees of freedom, but the pivot and wrist joints were powered using a belted bevel gear differential driven by two servos mounted in the arm.

This design was chosen initially because it would allow for 2 servos worth of power in each joint due to the differential, but during prototyping, the added complexity caused major backlash issues, which prevented accurate control.



The two servos are mounted inside the arm, and are offset to both fit inside the tight space. The power is delivered to the differential by GT2 Timing belts on a 1:1 ratio.

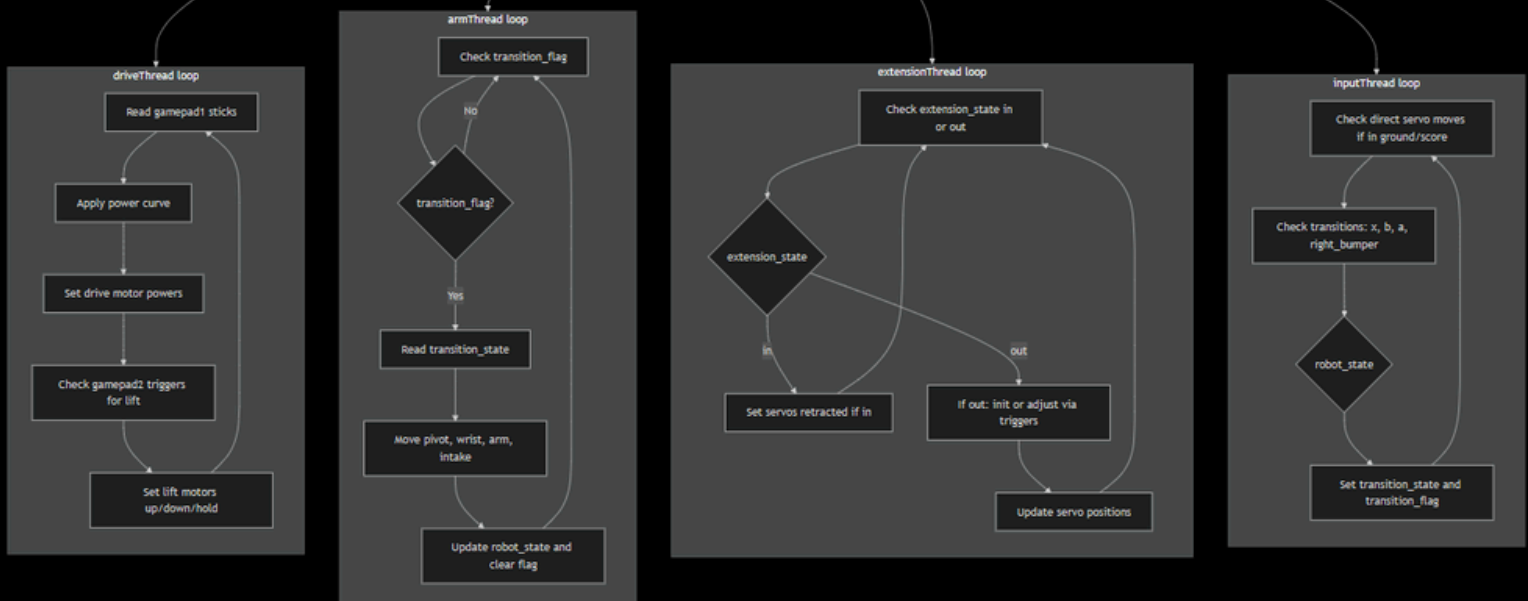
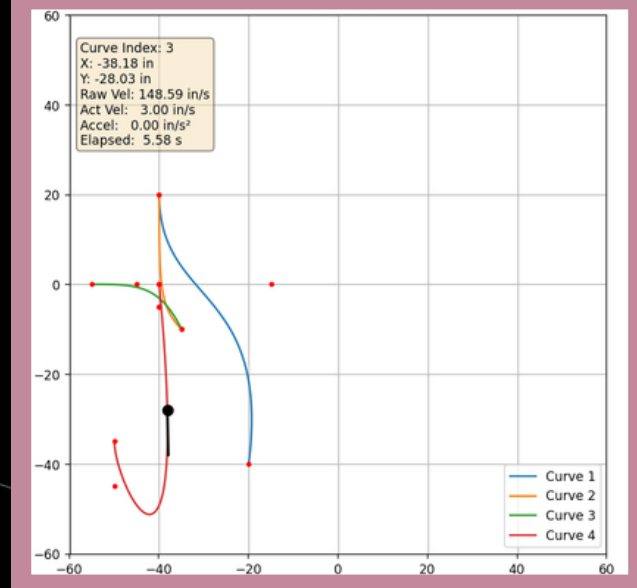
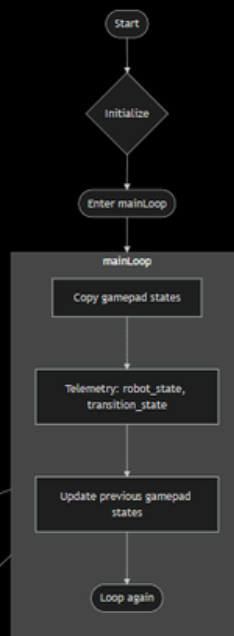
Brainstorming



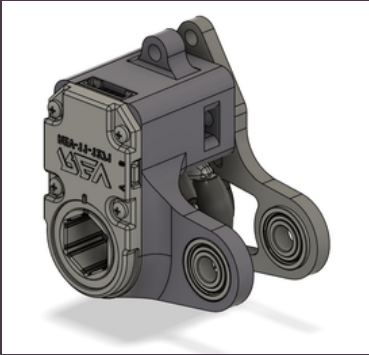
Autonomous Pathing

For autonomous control of the robot, we chose to use the PedroPathing Motion Library. This library uses a 3 odometry pod localization system, which allows for the robot to follow predefined paths. We use a pathing system rather than individual move commands because it allows for more efficient movements to save time and allow for higher scoring autos.

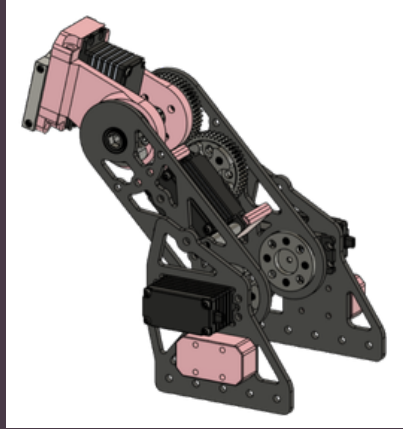
Our flow chart for the teleoperated control period demonstrates our automations within driver control. We have developed a simple control system that automates as many processes as possible, while giving the driver sufficient room for error. The code consists of 3 different states controlled via a state machine. An intaking state, an idle state, and a scoring state. Benefiting from our multi-threaded driver control, switching between these states automatically manipulates the sample/specimen from ground to scoring position, avoiding collisions and rotating the specimen for scoring. Toggle positions within each state allows for repeatability and consistency.



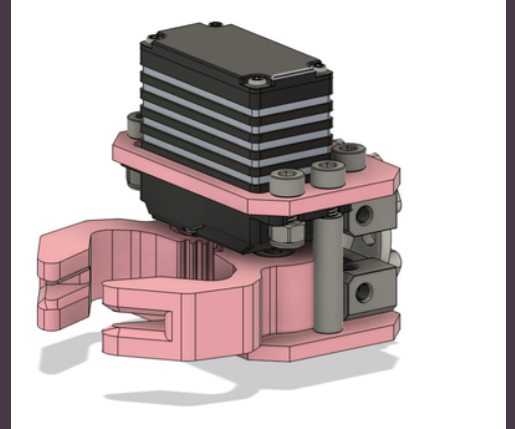
Current Robot



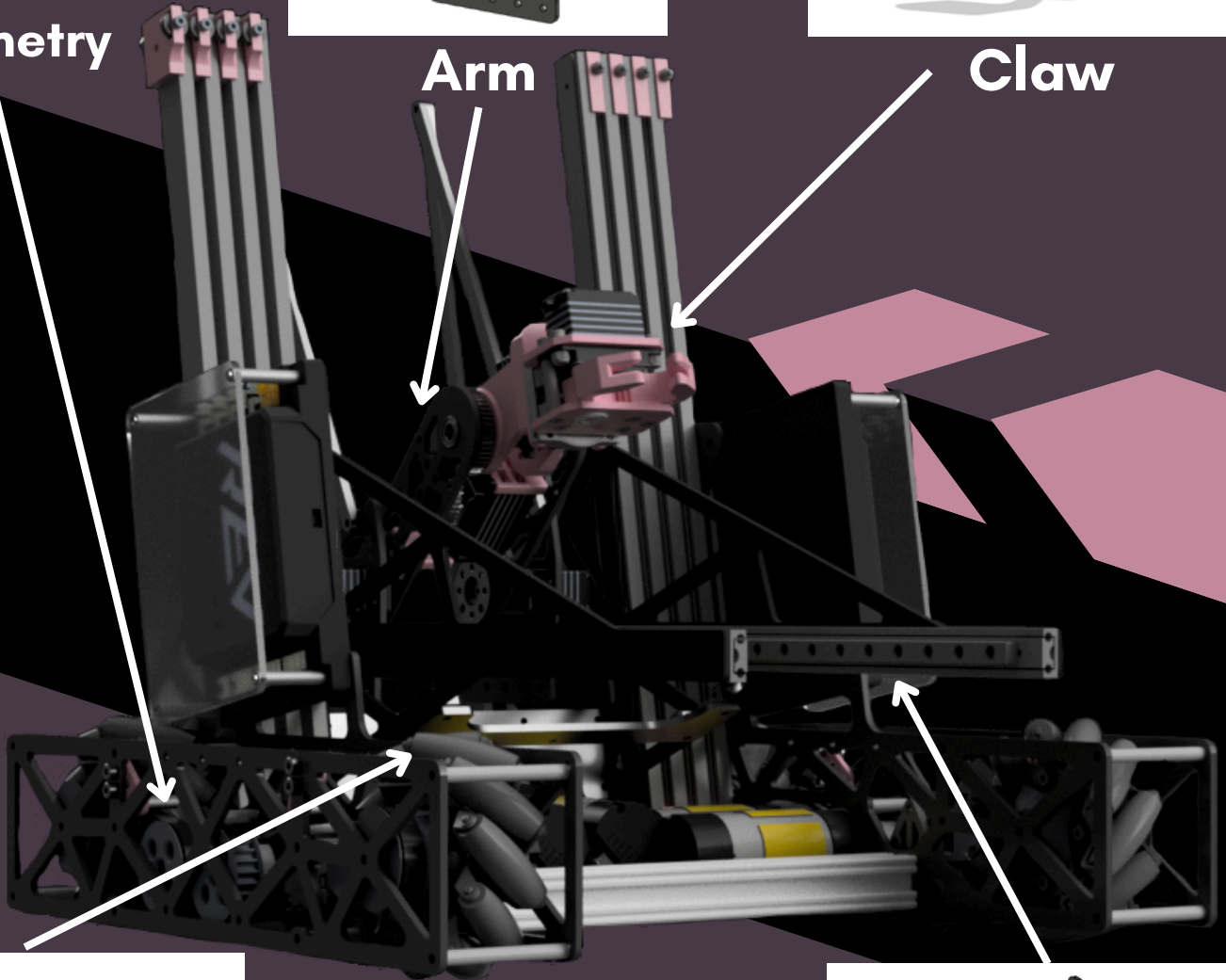
Odometry



Arm



Claw

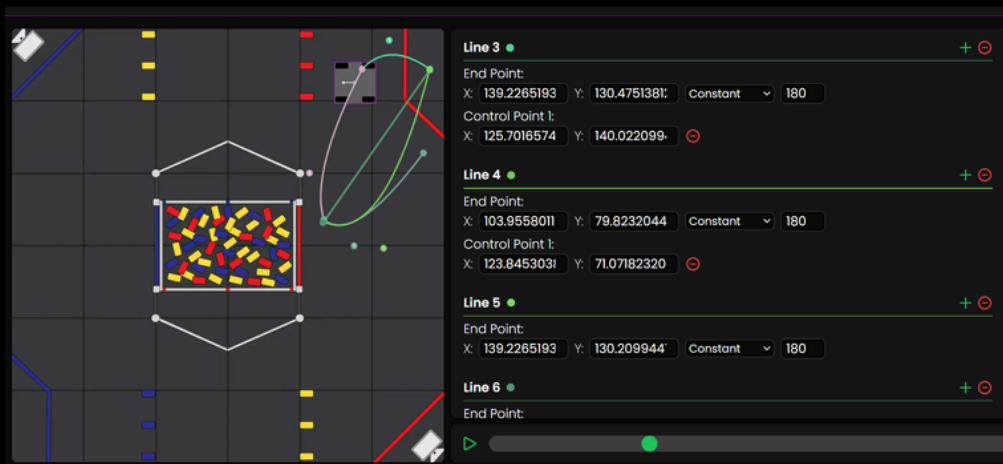


Lift

- #1 Ranked in SoCal League E
- 63pt Three-Specimen Auto
- 67pt Four-Sample Auto
- 10 Sample Average



Extension



ΣΕΒ

375